

# **GymHub**

## **A Product management Case Study**

### ***Product Overview***

**Product name:-** GymHub

#### **Tagline**

A smarter way to manage campus fitness.

#### **Product Vision**

GymHub is a mobile application proposed to improve the gym experience at **IIT Ropar** by enabling slot booking, equipment availability tracking, workout management, and streamlined communication between gym users and coaches.

#### **Problem Statement**

Campus gyms often experience overcrowding, inefficient equipment usage, and limited communication between students and coaches. Students frequently wait for equipment, are unaware of gym occupancy, and lack a centralized platform for updates and workout guidance. GymHub aims to address these challenges through a user-centric mobile application.

#### **Target Users**

##### **Primary Users**

- Students actively using the campus gym
- First-time gym users

##### **Secondary Users**

- Fitness Coaches
- Gym Administrators

#### **Product Goals**

- Reduce waiting time during peak hours.
- Enable gym slot booking.
- Improve equipment availability visibility.
- Simplify communication between coaches and students.
- Help users track workouts and progress.

# User Research

## Research Objective

To understand the daily challenges faced by campus gym users and identify opportunities to improve the overall gym experience through a digital product.

## Research Methodology

The research was conducted through direct observation of campus gym operations and informal discussions with regular gym users. The study focused on understanding recurring challenges related to gym accessibility, equipment usage, communication, and workout management. The findings were used to identify user pain points and define product requirements.

## Key Findings

| Objective                                                      | Impact                                               |
|----------------------------------------------------------------|------------------------------------------------------|
| Peak-hour overcrowding leads to waiting times.                 | Reduces workout efficiency and user satisfaction.    |
| Users cannot determine equipment availability before visiting. | Causes unnecessary waiting and congestion.           |
| No centralized platform for announcements or updates.          | Important information is often missed.               |
| Beginners struggle to find structured workout plans.           | Increases confusion and discourages consistency.     |
| Workout progress is tracked manually, if at all.               | Makes it difficult for users to monitor improvement. |

## Key Insights

- Peak-hour congestion significantly impacts the workout experience.
- Users value convenience and would benefit from visibility into gym occupancy.
- A centralized communication channel can reduce missed announcements.
- Beginner-friendly guidance can improve user engagement and retention.
- The identified pain points can be addressed through a digital platform focused on booking, communication, and workout management.

# ***User Personas***

## **Persona 1 : Srajan Srijariya - Regular Gym User**

**Age:** 19 | **Role:** B.Tech Student (3rd Year) | **Experience:** Intermediate

### **Goals**

- Complete workouts efficiently after classes.
- Improve strength and athletic performance.

### **Pain Points**

- Long waiting times for popular equipment.
- Overcrowding during peak hours.
- No way to check equipment availability before visiting.

### **Needs**

- Slot booking.
- Equipment availability.
- Workout tracking and personal records.

## **Persona 2 : Abhinav Kumar Singh - First Time Gym User**

**Age:** 19 | **Role:** B.Tech Student (2nd Year) | **Experience:** Beginner

### **Goals**

- Improve overall fitness.
- Learn correct exercise techniques.
- Build a consistent workout routine.

### **Pain Points**

- Doesn't know where to begin.
- Hesitant to ask experienced members for help.
- Unsure about proper workout planning.

### **Needs**

- Guided workout plans.
- Exercise tutorials.
- Progress tracking and reminders.

### **Persona 3 : Rahul Rohilla - Weightlifter**

**Age:** 28 | **Role:** PhD (4th year) | **Experience:** Advanced

#### **Goals**

- Optimize training for competitions.
- Track performance and personal records.
- Access equipment with minimal waiting.

#### **Pain Points**

- Peak-hour crowd disrupts training.
- No structured way to monitor progress.
- Competition announcements are scattered across different platforms.

#### **Needs**

- Personal record tracking.
- Competition and event notifications.
- Priority slot booking for training sessions (if applicable).

### **Person 4: Harinder Singh - Weightlifting & Gym Coach**

**Age:** 35 | **Role:** Weightlifting & Gym Coach

#### **Goals**

- Ensure smooth gym operations.
- Guide students effectively.
- Manage gym schedules and communication.

#### **Pain Points**

- Repeating the same announcements to different groups.
- Difficulty managing overcrowding.
- No centralized platform for attendance or communication.

#### **Needs**

- Dashboard to manage bookings.
- Attendance records.
- Announcement system.
- Ability to share workout plans and schedules.

# User Journey Mapping

## Current User Journey

| Stage         | User Action                                     | Pain Point                             |
|---------------|-------------------------------------------------|----------------------------------------|
| Before Visit  | Decides to go to the gym                        | Doesn't know if the gym is crowded.    |
| Arrival       | Reaches Gym                                     | May find long queues for equipment.    |
| Workout       | Waits for equipment before starting the workout | Time lost due to overcrowding.         |
| Communication | Checks WhatsApp/ Mails                          | Important announcements can be missed. |
| After Workout | Leaves the gym                                  | No way to track history or progress.   |

## Future User Journey (Using GymHub)

| Stage         | User Action                     | Improvement                                              |
|---------------|---------------------------------|----------------------------------------------------------|
| Before Visit  | Opens GymHub                    | Checks gym occupancy and available slots.                |
| Planning      | Books a workout slot            | Avoids overcrowding and long waiting times.              |
| Workout       | Uses available equipment        | Better time management and smoother workouts.            |
| Communication | Receives in-app notifications   | Never misses important announcements or schedule changes |
| After Workout | Logs workout and views progress | Tracks performance and stays motivated.                  |

## Key Opportunities

- Improve gym accessibility through slot booking.
- Reduce waiting time with occupancy visibility.
- Centralize communication between coaches and users.
- Help beginners through structured workout guidance.
- Enable users to track workouts and personal progress.

## Solution Ideation

Based on the user research and journey mapping, the following features were identified to address the major pain points faced by campus gym users and coaches.

### Student Mobile App

| Feature                | Purpose                                        |
|------------------------|------------------------------------------------|
| Slot Booking           | Reserve a workout slot to reduce overcrowding. |
| Gym Occupancy          | View real-time crowd levels before visiting.   |
| Equipment availability | Check the availability of key equipment.       |
| Workout Tracker        | Log workouts and monitor progress.             |
| Personal Records       | Record and track lifting milestones.           |
| Coach Announcements    | Receive important updates and notices.         |
| Event Registration     | Register for competitions and fitness events.  |
| Notifications          | Receive reminders and schedule updates.        |

### Coach Dashboard

| Feature             | Purpose                                        |
|---------------------|------------------------------------------------|
| Slot Management     | Manage booking capacity and schedules.         |
| Attendance Tracking | Monitor student attendance.                    |
| Equipment Status    | Update equipment availability and maintenance. |
| Announcement Portal | Broadcast notices to all gym users.            |
| Member Management   | View registered users and booking history.     |
| User Analytics      | Analyze peak hours and facility utilization.   |

## Feature Mapping

| User Pain Point         | Proposed Feature       |
|-------------------------|------------------------|
| Overcrowding            | Slot Booking           |
| Waiting for Equipment   | Equipment Availability |
| Missed Announcements    | Coach Announcements    |
| Lack of Workout Tracing | Workout Tracker        |
| Beginner Need Guidance  | Beginner Workout Plans |
| No Performance Records  | Personal Records       |

## Design Principles

- Simplicity – Keep the interface intuitive and easy to navigate.
- Accessibility – Enable quick access to essential features.
- Efficiency – Reduce waiting time and improve gym utilization.
- Transparency – Provide real-time visibility into occupancy and equipment.
- User-Centricity – Design around the needs of students and coaches.

# Feature Prioritization

To ensure a focused MVP and efficient development process, features were prioritized using the **MoSCoW prioritization framework**, categorizing them into Must Have, Should Have, Could Have, and Won't Have.

## MoSCoW Prioritization

| Must Have              | Should Have        | Could Have              | Won't Have                 |
|------------------------|--------------------|-------------------------|----------------------------|
| Slot Booking           | Workout Tracker    | Nutrition Tips          | AI Workout Recommendations |
| Gym Occupancy          | Personal Records   | Progress Photos         | Smartwatch Integration     |
| Equipment availability | Event Registration | Achievement badges      | Social Feed                |
| Coach Announcements    | Attendance History | Exercise Tutorials      | Diet planner               |
| Notifications          | Workout Plans      | Competition Leaderboard | Wearable Device Sync       |

## Why these priorities?

### Must Have

These features directly address the primary pain points identified during user research and are essential for delivering the core value of GymHub.

### Should Have

These features enhance the user experience but are not critical for the initial launch.

### Could Have

These features provide additional value and engagement but can be introduced in later iterations.

### Won't Have

These features are intentionally excluded from the first release to maintain focus on solving the most critical user problems.



## Minimum Viable Product (MVP)

Based on the feature prioritization, the MVP focuses on solving the most critical pain points identified during user research while keeping the initial product simple and practical to implement.

### MVP Features

| Feature                | Reason for Inclusion                               |
|------------------------|----------------------------------------------------|
| Slot Booking           | Reduces overcrowding and improves gym scheduling   |
| Live Gym Occupancy     | Allows users to check crowd levels before visiting |
| Equipment Availability | Minimizes waiting time for equipment.              |
| Coach Announcements    | Centralizes communication and important updates.   |
| Notifications          | Reminds users about bookings and announcements     |

### Features Deferred to Future Releases

| Feature            | Reason                                                         |
|--------------------|----------------------------------------------------------------|
| Workout Tracker    | Valuable but not essential for solving the primary problem.    |
| Personal Records   | Can be introduced after core operations are stable.            |
| Event Registration | Useful for engagement but not required in the first release    |
| Nutrition Tips     | Nice-to-have feature with lower immediate impact               |
| Exercise Tutorials | Planned for a later version after validating the core product. |

### MVP Success Criteria

- Reduce waiting time during peak hours.

- Improve utilization of gym facilities.
- Provide a centralized communication platform.
- Enable users to plan gym visits more efficiently.
- Validate demand before expanding feature offerings.

## Rice Prioritization

The RICE framework was used to prioritize MVP features based on four parameters: **Reach**, **Impact**, **Confidence**, and **Effort**. This framework helps identify features that deliver the highest value with the available development effort.

### What is RICE?

| Parameter  | Description                                           |
|------------|-------------------------------------------------------|
| Reach      | Number of users expected to benefit from the feature. |
| Impact     | Expected improvement in user experience.              |
| Confidence | Confidence in the assumptions based on research.      |
| Effort     | Estimated development effort required.                |

### Feature Evaluation

| Feature                | Reach | Impact | Confidence | Effort | RICE Score |
|------------------------|-------|--------|------------|--------|------------|
| Slot Booking           | 9     | 9      | 9          | 4      | 182        |
| Live Gym Occupancy     | 8     | 8      | 8          | 3      | 171        |
| Coach Announcements    | 9     | 7      | 9          | 2      | 284        |
| Equipment Availability | 7     | 8      | 7          | 5      | 78         |
| Notifications          | 8     | 6      | 8          | 2      | 192        |

**Note:** For simplicity, the RICE score is calculated as **(Reach x Impact x Confidence) / Effort**

# Insights

- Coach Announcements received the highest RICE score due to its high reach and low implementation effort.
- Slot Booking remains a core feature because it directly addresses the primary user pain point.
- Live Gym Occupancy provides significant value with moderate implementation complexity.
- Equipment Availability is valuable but requires additional operational support, making it a lower initial priority.
- Notifications complement other MVP features by improving user engagement and communication.

## Product Specifications

### Product Scope

The initial version of GymHub will provide slot booking, live gym occupancy, coach announcements, notifications, and a basic dashboard for gym coaches to manage bookings and communicate with users.

### Functional Requirements

| ID    | Requirement                           |
|-------|---------------------------------------|
| FR-01 | Students can book gym slots.          |
| FR-02 | Students can view live gym occupancy. |
| FR-03 | Students receive coach announcements. |
| FR-04 | Students receive notifications.       |
| FR-05 | Students can track workouts.          |
| FR-06 | Coaches can post announcements.       |
| FR-07 | Coaches can manage slot availability. |

## Non-Functional Requirements

| Requirement | Description                              |
|-------------|------------------------------------------|
| Performance | The app should load within 2 seconds.    |
| Usability   | Simple and intuitive interface.          |
| Reliability | Booking data should remain accurate.     |
| Scalability | Should support future feature additions. |
| Security    | User accounts should be authenticated.   |

## Assumptions

- Students have institute login credentials.
- Internet connectivity is available.
- Coaches regularly update announcements.
- Gym slot timings are predefined.

## Constraints

- Limited development resources.
- Initial rollout limited to IIT Ropar.
- Live occupancy depends on data updates.
- MVP excludes wearable integration and AI features.

## Out of Scope

- Diet planning
- AI workout recommendations
- Smartwatch integration
- Social networking
- Online coaching

## Conclusion

GymHub proposes a practical solution to improve the IIT Ropar campus gym experience by addressing challenges such as overcrowding, fragmented communication, and inefficient gym usage. Through user research, feature prioritization, MVP planning, and product specification, the case study demonstrates a structured product management approach to designing a feasible digital solution. Future iterations can expand the platform with advanced fitness tracking and analytics based on user feedback.